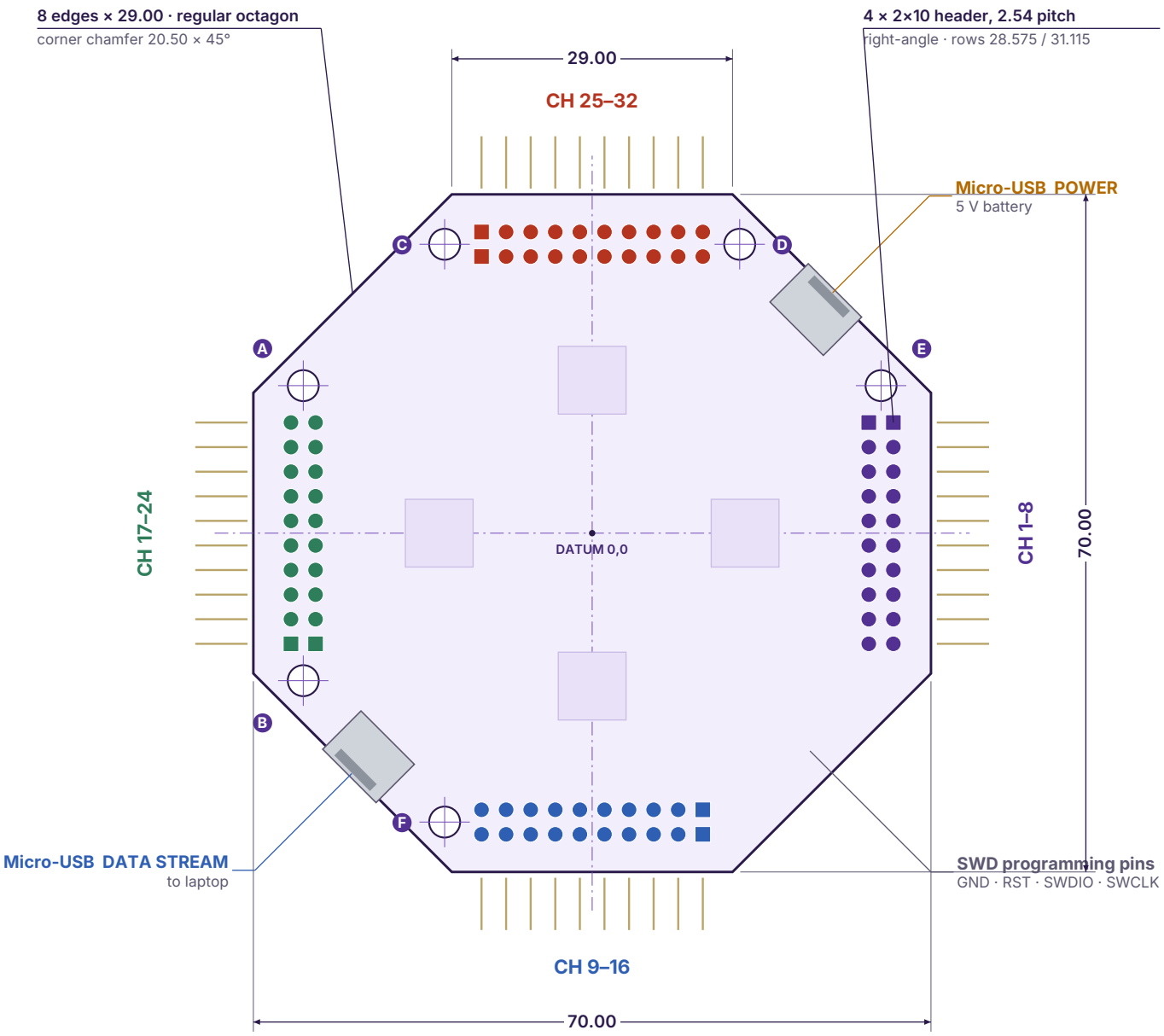


Mounting holes

Hole	X	Y	Ø	Type
A	-29.845	+15.240	3.20	NPTH, M3
B	-29.845	-15.240	3.20	NPTH, M3
C	-15.240	+29.845	3.20	NPTH, M3
D	+15.240	+29.845	3.20	NPTH, M3
E	+29.845	+15.240	3.20	NPTH, M3
F	-15.240	-29.845	3.20	NPTH, M3

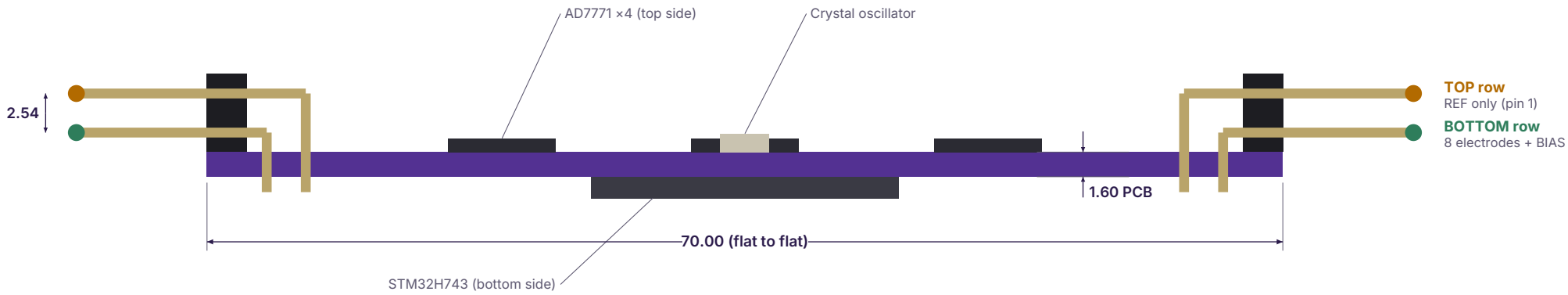
Notes

- Board outline: regular octagon, 70.00 across flats, 8 edges of 29.00 (corner chamfer 20.50 × 45°).
- Thickness 1.60 FR-4, 4 copper layers.
- Header pin rows at 28.575 (inner) and 31.115 (outer) from the datum; 10 pins at 2.54 pitch centred on each flat (±11.43).
- Right-angle pin tails extend beyond the board edge; allow clearance for mating 2.54 mm sockets.
- Corner positions of USB connectors are shown for orientation; body outlines are representative.
- Hole positions are taken directly from the NPTH drill file.

Title	IronBCI-32 — board outline & interfaces
Doc. No.	IRB32-MD-002
Revision	Rev. A · 10 Nov 2026
Units	Millimetres; datum 0,0 = board centre
Scale	1.5 : 1 (A4 print)
Source	KiCad 9.0.6 Gerber/drill data, Nov 2025

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02 SIDE ELEVATION AND HEADER DETAIL



Schematic elevation along the Y axis — component and header heights are illustrative, not to scale. Row positions (28.575 / 31.115) and PCB thickness are from fabrication data.

Parameter	Value
Header type	2 × 10, 2.54 mm pitch, right-angle
Row pitch	2.54
Pin span per row	22.86 (9 × 2.54)
Inner row	28.575 from datum
Outer row	31.115 from datum
Pin drill	Ø1.00 plated
Quantity	4 (one per AD7771)

Front view — looking into the CH 1–8 connector (chips up)

CH 1–8 header of AD7771 #1

TOP row	AVSS do not use	ref leave empty	ref leave empty	ref leave empty	ref leave empty	ref leave empty	ref leave empty	ref leave empty	ref leave empty	REF pin 1
BOTTOM	AVDD do not use	8 CP3	7 P3	6 O1	5 P7	1 F7	2 FT7	3 T7	4 TP7	BIAS pin 2

← outer end (power pins) pin-1 end →

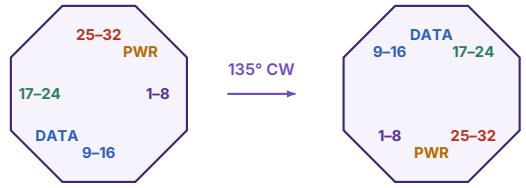
Same geometry on all four headers. Never connect anything to the AVSS / AVDD pins at the outer end. Full pin assignment: Connector Pinout & Electrode Guide (IRB32-PIN-004).

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03 INTERFACE LOCATIONS

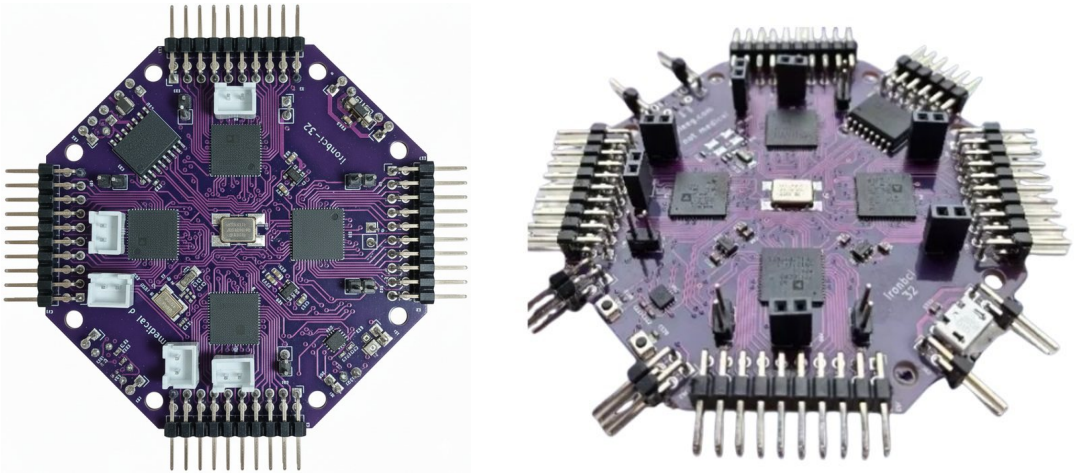
Interface	Edge (drawing view)	Position from datum	Function
CH 1–8	+X flat	rows at 28.575 / 31.115, centred	ADC #1 · REF + BIAS ear clips
CH 9–16	−Y flat	rows at 28.575 / 31.115, centred	ADC #2
CH 17–24	−X flat	rows at 28.575 / 31.115, centred	ADC #3
CH 25–32	+Y flat	rows at 28.575 / 31.115, centred	ADC #4
Micro-USB "Power"	+X / +Y diagonal edge	≈ (+21.8, +21.8), facing 45°	5 V input from USB battery only
Micro-USB "Data Stream"	−X / −Y diagonal edge	≈ (−21.8, −21.8), facing 225°	USB CDC serial to laptop
SWD pins	+X / −Y diagonal edge	—	ST-Link: GND, RST, SWDIO, SWCLK

Orientation key



Fabrication / drawing view — Pinout / user-guide view

This drawing uses fabrication orientation (as the Gerber files). User guides show the board rotated 135° clockwise, with Data Stream at the top and Power at the bottom.



Top view of the assembled board (left) and three-quarter view (right).

NOTE Enclosure.

3D-printable cases are published on Thingiverse (PiEEG/designs). For a custom case keep the six Ø3.2 holes free for M3 stand-offs and leave access to both Micro-USB ports, the SWD corner and the header pin tails.

CAUTION Handling.

The board has no case. Handle it by the edges, keep liquids away and avoid touching the analog front end.

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